# SAS Innovate 2026: PROC SUPERLEARNER MODEL

## Why use PROC SUPERLEANER ?

Machine learning is a branch of artificial intelligence that allows computers to learn from data through iterations. Different types of algorithms are trained on large datasets, and the system then uses data to make predictions, identify patterns, and make decisions leading to improved business performance over time. This introduces a new procedure in SAS Viya called PROC SUPERLEARNER. This procedure constructs a predictive model that combines multiple predictions from machine learning models.

## Demo Overview (~45 mins demo)

## Demonstration Steps

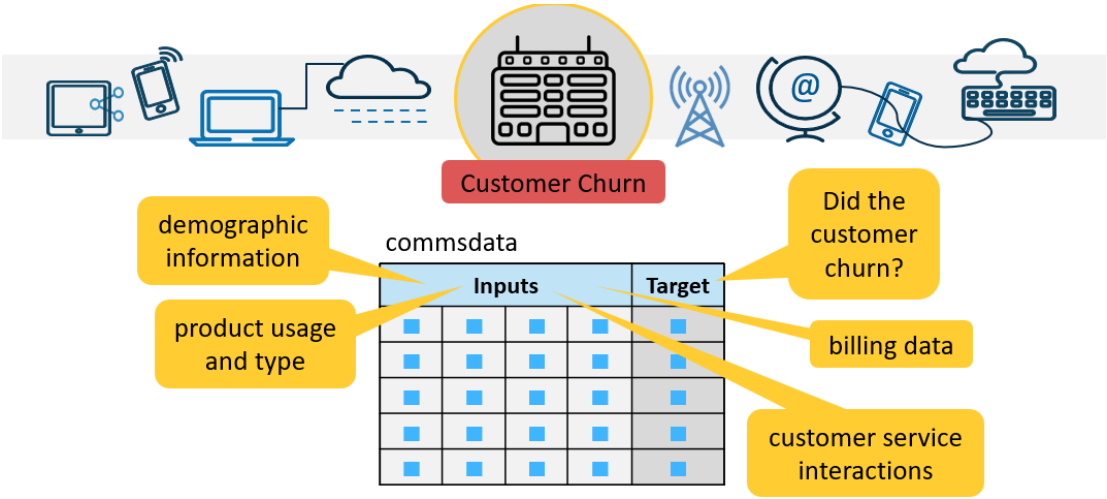
1. From the ribbon at the bottom of the screen, launch Google Chrome. When the browser opens, select **SAS Landing** from the bookmarks bar or from the link on the page.
2. If the user ID and password are not pre-filled, do the following: Enter **student** in the **User ID** field. Enter **Metadata0** in the **Password** field.
   - **Note**: Use caution when you enter the user ID and password because values can be case sensitive.
3. Click **Sign In**.
4. In the Assumable Groups window, select **Yes**. The SAS Landing home page appears.
5. From **SAS Landing**, you can access SAS Viya products.
   In the upper left corner click the **Applications** menu and select **SAS Studio: Develop Code and Flow**.
6. On the right of the interface navigate to the SAS Server tab, click on the tab.
   Navigate to the **"Files"** directory, under the File directory click **"Courses"** folder and navigate to the **"PROC_SUPERLEARNER"** folder.
7. Open the **PREPROCESSING_SUPERLEARNER.sas** file. **Run lines 15-86 ONLY,** this will handle the data preprocessing and formatting of our data. You may take time in between the demo model to run the bonus material.
8. Once the code has successfully executed, Open the SAS file labeled as **"PROC SUPERLEARNER MODEL.sas."**

**Folder Path :** Files → Home → Courses → PROC_SUPERLEARNER
Preprocessing_SUPERLEARNER.sas file (Run lines 15-86) → PROC SUPERLEARNER MODEL.sas

## Data Preparation

In this demonstration, we will create a new project in Model Studio. This is the project that is used throughout this 45-minute demonstration. During this demo, we will load the **Comms** dataset into memory from a local drive in the virtual lab environment.

Demo Scenario:
A fictitious telecommunications company has a problem with customer's churn. When existing customers or subscribers stop doing business or otherwise end a relationship with a company, this is known as churn, and this is bad for the bottom line. It is more costly to attract new customers than to keep existing customers. The data contains information about customer behavior, it consists of demographic information, product usages, product types, billing data, and customer service interactions.




In this stage we will explore the data and give some high-level overview of the data. We first must select a target variable. Target in this case will be the "BAD". Whether the customer defaulted on their loan or not. This is binary target, which gives us a sense of direction of how to start this process.

## Load data In-Memory

In this section, we will load our "COMMS" dataset into our SAS Viya environment using the LIBNAME statement. This statement allows us to connect a user-defined shortcut to a physical folder on our computers. It points to an existing directory to make dataset within it directly accessible in SAS.

```
libname ps '/home/student/Courses/PROC_SUPERLEARNER/data';
libname out '/home/student/Courses/PROC_SUPERLEARNER/output';
```

We create to LIBNAME statement "ps" & "out", this is where we will store our data that we load into our environment and where we store our output datasets.

## Data STEP: Loading data into CAS Server

```
/***    LOAD DATA IN-MEMORY     *******/
data ps.telecomms;
    set ps.commsdata;
run;
```

In this code, we name our new data table as "**ps.telecomms"** . We call our data telecommunication dataset from our data file location as **"ps.commsdata"**.

## Data Exploration

Explore by using PROC DESCRIBE procedure to provide a summary statistic of the data. Provide the shape of the data, check for missing data and the distribution of data as well.

```
proc means data= ps.telecomms N Mean std min max skewness kurtosis;
run;
```

The PROC MEANS procedure provides a data summarization tool to compute descriptive statistics for variables across all observations and within groups of observations. PROC MEANS does the following:

- Calculates descriptive statistics based on moments
- Estimate quantiles, which includes the median
- Calculates confidence limits for the mean
- Identifies extreme values
- Performs a *t* test

## Feature Engineering / Data Preprocessing

In this section, we will handle some feature engineering, from what we gather so far from our summary statistic is that within the data we have some variables that have negative minimum values for our customer data. It is possible that for some industry, the negative values are used to indicate account credit for a customer. For our demonstration we will handle the negative values  by setting all negative values ( < 0) to zero. By removing the negative values from our data it will ensure that we have no outliers that could drastically affect our predictions.

```
/**  Handling the negative values **/;
data ps.tele_comms;
    set ps.telecomms;

 /**  The original data with negative values   **/;
 array old_limits(*) tot_mb_data_roam_curr seconds_of_data_norm lifetime_value
bill_data_usg_m03
 bill_data_usg_m06 voice_tot_bill_mou_curr mb_data_usg_roamm01 mb_data_usg_roamm02
mb_data_usg_roamm03
 mb_data_usg_m01 mb_data_usg_m02 mb_data_usg_m03 calls_total calls_in_pk calls_out_pk
calls_in_offpk calls_out_offpk
 mb_data_ndist_mo6m data_device_age mou_onnet_pct_MOM mou_total_pct_MOM;

 array new_limits(*) tot_mb_data_roam_curr seconds_of_data_norm lifetime_value
bill_data_usg_m03
 bill_data_usg_m06 voice_tot_bill_mou_curr mb_data_usg_roamm01 mb_data_usg_roamm02
mb_data_usg_roamm03
 mb_data_usg_m01 mb_data_usg_m02 mb_data_usg_m03 calls_total calls_in_pk calls_out_pk
calls_in_offpk calls_out_offpk
```

```
mb_data_ndist_mo6m data_device_age mou_onnet_pct_MOM mou_total_pct_MOM;

/**  Apply rule to set negative values to zero  **/;
do i=1 to dim(old_limits);

/**  If the old value is a negative value then we will set the new value to 0  **/

   if old_limits[i] < 0 then new_limits[i] = 0;
   /** If the value is greater than 0, then we keep the original value **/
   else old_limits[i] = new_limits[i];
   end;
run;
```

The MEANS Procedure

| Variable | Label | N | Mean | Std Dev | Minimum | Maximum | Skewness | Kurtosis |
|---|---|---|---|---|---|---|---|---|
| Customer_ID | Primary Key | 56557 | 1871721.32 | 1214488.00 | 471.0000000 | 3999922.00 | 0.0549030 | -1.2549491 |
| churn | Churn Flag | 56557 | 0.1213289 | 0.3265120 | 0 | 1.0000000 | 2.3195744 | 3.3805450 |
| lifetime_value | Lifetime Value | 56557 | 5281.53 | 5068.84 | -14006.00 | 60740.20 | 1.8543235 | 5.7694165 |
| avg_arpu_3m | 3M Avg Revenue per User | 55437 | 60.2948845 | 22.8771098 | 0 | 160.3761848 | 0.8185036 | 2.3206915 |
| acct_age | Account Tenure | 56557 | 45.1726555 | 12.9064363 | 18.0000000 | 165.0000000 | 0.6693952 | 1.7918378 |
| billing_cycle | Billing Cycle | 56557 | 6.6233357 | 3.1898376 | 1.0000000 | 12.0000000 | -0.0289219 | -0.8397392 |
| nbr_contracts_ltd | Total Number Contracts Lifetime | 56557 | 4.1616747 | 2.6373436 | 1.0000000 | 16.6453080 | 0.9679281 | 1.2641342 |
| rfm_score | Account Ranking (RFM Score) | 56557 | 221.9940768 | 82.0648094 | 111.0000000 | 333.0000000 | 0.000131570 | -1.4704193 |
| Est_HH_Income | Estimated HH Income | 56557 | 31734.13 | 13799.92 | 0 | 263400.00 | 1.4026823 | 7.9566850 |
| cs_med_home_value | Census Area Median Home Value Index | 56434 | 2.1796190 | 1.5152938 | 0 | 9.9900000 | 1.9978241 | 5.5990919 |
| cs_pct_home_owner | Census Area Percent Home Owner | 56434 | 0.5777558 | 0.2602454 | 0 | 0.9900000 | -0.4732351 | -0.7986969 |
| cs_ttl_pop | Census Area Total Population | 55239 | 27601.26 | 18987.27 | 7.0000000 | 114124.00 | 1.0247724 | 1.4580546 |
| cs_hispanic | Census Area Hispanic | 55169 | 12.7517981 | 18.8474017 | 0.0400000 | 98.9100000 | 2.2799542 | 4.9767429 |
| cs_caucasian | Census Area Caucasian | 55235 | 69.6051136 | 27.2503133 | 0.3900000 | 99.8700000 | -1.0102982 | -0.0922614 |
| cs_afr_amer | Census Area African-American | 54695 | 11.5994990 | 18.4501180 | 0.0100000 | 100.0000000 | 2.5168167 | 6.3704496 |
| cs_other | Census Area Other | 55222 | 6.1808817 | 8.7397736 | 0.0200000 | 99.2600000 | 4.6557528 | 29.8402548 |
| cs_ttl_urban | Census Area Total Urban | 55239 | 79.1021812 | 31.4095466 | 0 | 100.0000000 | -1.5721755 | 1.2131983 |
| cs_ttl_rural | Census Area Total Rural | 55239 | 20.8978213 | 31.4095475 | 0 | 100.0000000 | 1.5721752 | 1.2131974 |
| cs_ttl_male | Census Area Total Males | 55239 | 49.0903170 | 2.4321788 | 0.6900000 | 97.1300000 | 3.8606207 | 47.0060458 |
| cs_ttl_female | Census Area Total Female | 55239 | 50.9096834 | 2.4321785 | 2.8700000 | 99.3100000 | -3.8606221 | 47.0060668 |
| cs_ttl_hhlds | Census Area Total Households | 55232 | 10088.67 | 6623.03 | 2.0000000 | 61774.00 | 0.8334123 | 1.3085916 |
| cs_ttl_mdage | Census Area Median Age | 55239 | 35.3785568 | 5.1257309 | 14.0000000 | 75.0000000 | 0.7833713 | 4.9415544 |
| forecast_region | Forecasted Region Key | 56557 | 45.9955090 | 26.8456454 | 0 | 92.0000000 | 0.000147342 | -1.2004130 |
| mb_inclplan | Plan Data MB | 56557 | 849.8417526 | 439.6062976 | 250.0000000 | 1500.00 | 0.2337719 | -1.2799281 |
| ever_days_over_plan | Total Days Over Plan | 56557 | 13.7506586 | 15.8381629 | 0 | 142.0000000 | 1.6871174 | 4.1333637 |
| ever_times_over_plan | Total Times Over Plan | 56557 | 2.5303499 | 2.4527833 | 0 | 26.0000000 | 1.0691828 | 1.5822748 |
| data_device_age | Avg Age of Devices on Plan | 52835 | 924.1027463 | 617.5700425 | -50.0000000 | 2600.19 | 0.4496267 | -0.3694745 |
| equip_age | Handset Age | 56557 | 20.0226851 | 13.1547543 | 0 | 49.0000000 | -0.2059709 | -1.2566574 |
| mfg_apple | Own Apple | 56557 | 0.4552398 | 0.4979969 | 0 | 1.0000000 | 0.1797671 | -1.9677534 |
| mfg_samsung | Own Samsung | 56557 | 0.2319076 | 0.4220541 | 0 | 1.0000000 | 1.2704621 | -0.3859397 |
| mfg_htc | Own HTC | 56557 | 0.0884771 | 0.2839900 | 0 | 1.0000000 | 2.8982527 | 6.4000952 |
| mfg_motorola | Own Motorola | 56557 | 0.0439910 | 0.2050769 | 0 | 1.0000000 | 4.4473562 | 17.7796063 |
| mfg_lg | Own LG | 56557 | 0.1142387 | 0.3181038 | 0 | 1.0000000 | 2.4254655 | 3.8830202 |
| mfg_nokia | Own Nokia | 56557 | 0.0611242 | 0.2395601 | 0 | 1.0000000 | 3.6641445 | 11.4263587 |
| delinq_indicator | Delinquent Indicator | 56557 | 0.1265272 | 0.3883281 | 0 | 4.0000000 | 3.5727607 | 15.5104632 |
| times_delinq | Consecutive Mths Delinquent | 56557 | 0.3919232 | 0.7062139 | 0 | 6.0000000 | 1.8791267 | 3.2411619 |
| count_of_suspensions_6m | Times Suspended Last 6M | 56557 | 0.4426266 | 0.9443556 | 0 | 5.0000000 | 2.6743535 | 7.0331033 |
| avg_days_susp | Days Suspended Last 6M | 56557 | 3.4713735 | 3.8312731 | 0 | 62.0000000 | 1.5937222 | 5.0680748 |
| calls_total | Total Calls Curr | 56557 | 727.7480985 | 615.5984674 | -1837.35 | 7949.91 | 1.7104540 | 4.7739613 |
| calls_in_pk | Calls Incoming Peak | 56557 | 235.4593419 | 241.5422011 | -1816.00 | 3487.33 | 1.5665171 | 7.1118747 |
| calls_in_offpk | Calls Incoming Off-Peak | 56557 | 388.6468614 | 406.1405144 | -1410.35 | 4640.24 | 1.8185824 | 5.6826097 |
| calls_out_offpk | Calls Outgoing Off-Peak | 56557 | 31.3590419 | 30.1620496 | -109.0000000 | 331.0000000 | 1.3056653 | 3.8581909 |
| calls_out_pk | Calls Outgoing Peak | 56557 | 72.2828533 | 83.4204312 | -498.0000000 | 1603.67 | 2.5904642 | 14.5607265 |
| mou_total_pct_MOM | Minutes Total Pct Change Month over Month | 47453 | 0.0572678 | 1.8850544 | -29.0000000 | 92.3000000 | 3.4333704 | 150.4456332 |
| mou_onnet_pct_MOM | Minutes On Network Pct Change Month over Month | 29812 | -0.2595395 | 3.8853929 | -45.0000000 | 124.7272727 | 2.8059664 | 91.5615366 |
| mou_roam_pct_MOM | Minutes Roaming Pct Change Month over Month | 21245 | 0.3590290 | 10.9338966 | -83.0000000 | 319.7000000 | 2.7611304 | 71.8312286 |
| voice_tot_bill_mou_curr | Total Voice Billed Minutes of Use | 56557 | 5836.33 | 5639.84 | -10912.74 | 72726.56 | 2.1484758 | 7.3929925 |
| tot_voice_chrgs_curr | Total Voice Charges | 55437 | 34.4301344 | 11.7843762 | 0 | 93.8935079 | 0.3894804 | 0.5010001 |
| tot_drpd_pr1 | Number of Dropped Calls 1 Mth Prior | 50754 | 2.6227095 | 3.6448450 | 0 | 34.0000000 | 3.2757163 | 15.3655455 |
| bill_data_usg_m03 | 3M Avg Billed Data Usage | 56557 | 1864.91 | 1634.51 | -13678.00 | 40767.10 | 1.3974299 | 13.7883885 |
| bill_data_usg_m06 | 6M Avg Billed Data Usage | 56557 | 1869.14 | 1418.91 | -10874.30 | 16813.60 | 1.3000284 | 5.0747826 |
| bill_data_usg_m09 | 9M Avg Billed Data Usage | 56557 | 1919.54 | 1862.27 | 0 | 84386.50 | 5.6973783 | 113.3316954 |
| mb_data_usg_m01 | MB Data Usage 1 Mth Prior | 56557 | 1697.20 | 1549.21 | -2425.00 | 40568.65 | 2.0378909 | 15.4829147 |
| mb_data_usg_m02 | MB Data Usage 2 Mths Prior | 56557 | 1698.65 | 1547.89 | -2171.06 | 40761.34 | 2.0176794 | 15.3247900 |
| mb_data_usg_m03 | MB Data Usage 3 Mths Prior | 56557 | 1696.25 | 1556.10 | -1621.00 | 40784.23 | 2.0524790 | 15.6742987 |
| mb_data_ndist_mo6m | 6M Avg Billed Data Usage Normally Distributed | 56442 | 0.0626561 | 4.1430941 | -92.4717301 | 87.7572409 | 0.1290002 | 18.7767314 |
| mb_data_usg_roamm01 | MB Data Usage Roam 1 Mth Prior | 56557 | 133.8719876 | 573.2160210 | -4358.00 | 11400.00 | 1.2315465 | 12.1567053 |
| mb_data_usg_roamm02 | MB Data Usage Roam 2 Mths Prior | 56557 | 125.7287692 | 603.0714436 | -7033.00 | 18727.00 | 2.2428314 | 39.4702820 |
| mb_data_usg_roamm03 | MB Data Usage Roam 3 Mths Prior | 56557 | 135.2706827 | 649.9658573 | -5984.00 | 12720.00 | 1.3668832 | 16.5920467 |
| tot_mb_data_curr | Total MB of Data Usage | 56557 | 2361.74 | 2289.80 | -10786.47 | 66288.07 | 4.6153732 | 49.8621406 |
| tot_mb_data_roam_curr | Total MB of Roam Data Usage | 56557 | 477.4561657 | 2887.31 | -60034.48 | 66288.07 | 2.1106411 | 32.9572663 |
| bill_data_usg_tot | Total Billed Data Usage | 56557 | 44.6595470 | 11.0659616 | 17.0000000 | 82.0000000 | 0.4048293 | -0.1327397 |

We also noticed that from the Data Exploration, we saw high levels skewness beginning with MB_Data_Usg. We create a DATA step where we call the variables MB_Data_Usg_M04 through MB_Data_Usg_M09. We will handle this by taking the LOG of the of these six variables to ensure evenly distributed data.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| MB_Data_Usg_M04 | MB of Data Usage Month 4 | 56557 | 159.3068586 | 381.1479077 | 0 | 14606.00 | 9.8151533 | 179.7305071 |
| MB_Data_Usg_M05 | MB of Data Usage Month 5 | 56557 | 142.7952685 | 471.5577885 | 0 | 24707.00 | 13.6121741 | 315.4085581 |
| MB_Data_Usg_M06 | MB of Data Usage Month 6 | 56557 | 230.5485970 | 718.7863878 | 0 | 29676.00 | 15.2432110 | 360.4407283 |
| MB_Data_Usg_M07 | MB of Data Usage Month 7 | 56557 | 94.2739537 | 259.8390503 | 0 | 13672.00 | 17.6345171 | 495.6025665 |
| MB_Data_Usg_M08 | MB of Data Usage Month 8 | 56557 | 109.5912442 | 348.7336436 | 0 | 16297.00 | 16.9031377 | 467.9810534 |
| MB_Data_Usg_M09 | MB of Data Usage Month 9 | 55660 | 95.9823751 | 218.3546743 | 0 | 8869.00 | 7.8438804 | 131.2951273 |

```
data ps.telecomms;
    set ps.telecomms;
    /* Original variables */
    array old_vars(*) MB_Data_Usg_M04 MB_Data_Usg_M05 MB_Data_Usg_M06
        MB_Data_Usg_M07 MB_Data_Usg_M08 MB_Data_Usg_M09;

    /* New variables */
    array new_vars(*) LOG_MB_Data_Usg_M04 LOG_MB_Data_Usg_M05
        LOG_MB_Data_Usg_M06 LOG_MB_Data_Usg_M07 LOG_MB_Data_Usg_M08
        LOG_MB_Data_Usg_M09;

    /* Apply rule */
    do i=1 to dim(old_vars);
        /* If negative → log transform */
        if old_vars[i] < 0 then new_vars[i]=log(abs(old_vars[i]));

        /* If zero or positive → keep original value */
        else new_vars[i]=old_vars[i];
    end;

    /* Drop variables (outside loops) */
    drop i upsell_xsell city city_lat city_long data_usage_amt
        mou_onnet_6m_normal mou_roam_6m_normal region_lat region_long state_lat
        state_long tweedie_adjusted zip_lat zip_long zipcode_primary;
run;

data ps.tele_comms;
    set ps.telecomms;
```

```
    drop MB_Data_Usg_M04 MB_Data_Usg_M05 MB_Data_Usg_M06 MB_Data_Usg_M07
        MB_Data_Usg_M08 MB_Data_Usg_M09;
run;

/*********************************************************/
proc format;
    value churn 0='Stay' 1='Churn';
run;

data ps.telecomms;
    set ps.telecomms;
    format churn churn.;
run;
```

This section of code handles our extreme outliers where some of the values are negative numbers, we utilize array index on our original variables to create our new values to replace the values. The following data step handles the skewness; we take the log of our six variables to improve the distribution of the data. After we have successfully executed the DATA step, we will then drop the original MB_Data_Usg variables.

## Data Partition

The next step of this demonstration is to partition our data into our training and validation set of analysis; we utilize our target variable for the partition. Remember we are working with binary target variable with customers ending business with the fictitious telecommunication company.

```
/***** DATA PARTITION:SPLITTING OUR DATA INTO A 70/30 SPLIT  *************/;
proc  partition data=ps.telecomms samppct=70 seed=12345 partind;
    by churn;
    output out=ps.telecomms_part copyvars=(_all_);
run;
```

The PARTITION Procedure

| Stratified Sampling Frequency | | | |
|---|---|---|---|
| Index | Churn Flag | Number of Obs | Number of Samples |
| 0 | Stay | 49695 | 34787 |
| 1 | Churn | 6862 | 4803 |

| Output Tables | | | |
|---|---|---|---|
| Library | Name | Number of Rows | Number of Columns |
| OUT | TELECOMMS_PART | 56557 | 114 |

In the following code, we begin with the partitioned dataset created using the PROC PARTITION procedure. A random seed is specified, and the target variable churn is defined at an interval level. We split our data into 70 training and 30 percent validation, we implement a random seed, we specify we want to include the partition indicator within our dataset. The specficy our target variable of churn. We produce an output saved to our LIBREF "ps" labeled as telecoms_part. Finally, we copy over all variables and run the PROC PARTITION procedure.

## Format Churn

In this section we change the format of the variable churn, we do this by specifying whether a person stays or churns( ends business with the telecommunication company). The reason for this is that the model will only observe the probability of the lowest level of the TARGET alphanumerically. If we leave the churn outcomes as 1 and 0 the model will only observe the observations of a customer continuing business with the telecommunication company, but the goal is to provide predictive probabilities of a customer you have ended business with the telecommunications company.

```
/** Modify the Partition Indicator: 0 =Training,1 = Validation  ****/
proc format;
    value _partind_ 0='Training' 1='Validation';
run;

/**Applying format to the variable churn**/

data out.telecomms_part;
    set out.telecomms_part;
    format _partind_ _partind_.;
run;
```

We use the **PROC FORMAT** procedure we select the value that we want to format as "_partind_". We format **"0"** as Training and **"1"** as Validation. Next, we apply the format to the variable **"_PartInd_"** to include the new format.

## Macro-Variable Inputs\

In this section, we provide a list of input variables that we suggest using for the model. This input variable can vary, but we suggest using the listed inputs for simplicity of the demonstration.

```
/***  Macro-Variable for Inputs for Model  ****/
%let VI_inputs=pymts_late_ltd ever_days_over_plan mou_total_pct_MOM
      curr_days_susp  delinq_indicator  times_delinq  mou_roam_pct_MOM
     equip_age
    bill_data_usg_tot nbr_contracts_ltd rfm_score Est_HH_Income
       cs_med_home_value   cs_pct_home_owner   cs_ttl_pop   cs_hispanic
     cs_caucasian
      cs_afr_amer   cs_other   cs_ttl_urban   cs_ttl_rural   cs_ttl_male
     cs_ttl_female
    cs_ttl_hhlds cs_ttl_mdage forecast_region mb_inclplan;
```

The macro variable **&VI_inputs** are used to represent the input variables, which are specified at an interval level.

## Building the PROC SUPERLEARNER Model: BASELEARNER Algorithms

```
/***************BUILDING THE PROC SUPERLEARNER MODEL **********/;


proc superlearner data=out.telecomms_part(where=(_PartInd_= 1 ))
seed=23456;
    target churn / level=nominal;
    input &VI_inputs / level=interval;
    baselearner 'GB' gradboost(leafsize=50 numbin=100);
    baselearner 'Forest' forest( maxdepth=15 ntrees=100
inbagfraction=0.6
        minleafsize=5);
    baselearner 'DT' treesplit;
```

```
    baselearner 'Logistic' logselect(link=logit selection=stepwise);
    crossvalidation kfold=3;
    output out=ps.predict2_output2 copyvar=churn;
    store out=out.scoreData2;
run;
```

We build the remainder of the model using the BASELEARNER statements. In this model, we utilize the light gradient boosting, the Forest, the Decision Tree, and the Logistic Regression model. We use more baselearners and specify the hyperparameters for each of the models. By default, each baselearner has its own set of hyperparameters that you can adjust to optimize the model. The Cross-validation is performed using a 3-fold approach, and the prediction results are written to the output table out.predict_output. We use the store statement to save and create our score data labeled as out.scoreData2 to be used against new or existing holdout data for model accuracy.

The SUPERLEARNER Procedure

| Model Information | |
|---|---|
| Data Source | TELECOMMS_PART |
| Target Variable | churn |
| Target Variable Type | NOMINAL |
| Number of Folds | 3 |
| Number of Base Learners | 4 |
| Loss Function | Log Loss |
| Meta-learning Method | Convex-constrained Binomial Likelihood Maximization |
| Random Number Seed | 23456 |

| | |
|---|---|
| Number of Observations Read | 39590 |
| Number of Observations Used for Training | 14309 |
| Number of Observations Used for Scoring | 39590 |

Probability modeled is churn = Churn.

| Super Learner Model Coefficients | | | |
|---|---|---|---|
| Name | Model Type | Coefficient | Cross-Validated Risk |
| LGB | Light Gradient Boosting Machine | 0 | 0.31661 |
| Forest | Forest | 0.95481 | 0.20683 |
| DT | Decision Tree | 0 | 0.12654 |
| Logistic | Logistic Regression | 0.04519 | 0.22610 |

| Output Tables | | | |
|---|---|---|---|
| Library | Name | Number of Rows | Number of Columns |
| OUT | SCOREDATA2 | 1 | 2 |
| PS | PREDICT2_OUTPUT2 | 39590 | 2 |

In the above figure, The SUPERLEARNER procedure was applied to the TELECOMMS_PART dataset to predict the target variable churn using 3-fold cross-validation and **four base learners: light gradient boosting (LGB), the Forest, the Decision Tree (DT), and the Logistic Regression model)**. The ensemble used quadratic loss with a convex constrained least squares meta-learning method to combine model predictions. From the results of the SUPERLEARNER model coefficient, we see that the Light Gradient Boosting (LGB) did not contribute to the final ensemble, receiving the largest cross-validated risk of 0.31 and the Decision Tree with the second highest cross-validated risk of 0.22. The Forest model was the overall contributor with a coefficient of 0.95 with a cross validated risk of 0.12 and the same for Logistic Regression. The final predictions were saved to the **"PS.PREDICT_OUTPUT"** table.

## Score Output for Validation Data

```
/*******Score Validation from the Previous Model************/
proc superlearner data=out.telecomms_part restore=out.score_model;
    output out=out.scoreData2 learnerpred copyvars=(churn _partind_);
run;

proc print data=out.scoredata2(obs=10);
run;
```

| Scoring Information | |
|---|---|
| Store | SCORE_MODEL |
| Scoring Table | TELECOMMS_PART |
| Training Table | TELECOMMS_PART |
| Target Variable | churn |

| | |
|---|---|
| Number of Observations Read | 56557 |
| Number of Observations Used for Scoring | 21245 |

Probability modeled is churn = Churn.

| Output Tables | | | |
|---|---|---|---|
| Library | Name | Number of Rows | Number of Columns |
| OUT | SCOREDATA2 | 56557 | 6 |

| Obs | P_churnChurn | LGB | Forest | Logistic | churn | _PartInd_ |
|---|---|---|---|---|---|---|
| 1 | 0.10672 | 0.13723 | 0.11750 | 0.10151 | Stay | Training |
| 2 | 0.12226 | 0.21489 | 0.13931 | 0.11403 | Churn | Training |
| 3 | 0.23784 | 0.33300 | 0.41263 | 0.15348 | Stay | Training |
| 4 | 0.13883 | 0.05764 | 0.06998 | 0.17206 | Stay | Validation |
| 5 | 0.69125 | 0.52426 | 0.62931 | 0.72114 | Stay | Validation |
| 6 | 0.28641 | 0.07740 | 0.12394 | 0.36484 | Stay | Training |
| 7 | 0.30565 | 0.25151 | 0.18530 | 0.36374 | Stay | Validation |
| 8 | 0.13104 | 0.16170 | 0.12164 | 0.13558 | Stay | Validation |
| 9 | 0.09110 | 0.03948 | 0.04781 | 0.11200 | Stay | Training |
| 10 | 0.08534 | 0.04111 | 0.05022 | 0.10230 | Stay | Validation |

In this section of the code, we take score data from the PROC SUPERLEARNER model, to use for comparison on our holdout or validation set to test the model accuracy. We call the PROC SUPERLEARNER statement, but we will use the procedure restore to call the score data from the PROC SUPERLEARNER Model. We create and output table labeled as **"OUT.SCOREDATA2"**

## Assessing The Event of Churn

```
/**** ASSESSING EVENT CHURN by the Partition  ***/;
ods output ROCInfo=out.roc_table2;

/* <----- This is the ROC Curve table output */
proc assess data=out.scoredata2;
    var P_churnChurn;
    target churn / event="Churn" level=nominal;
```

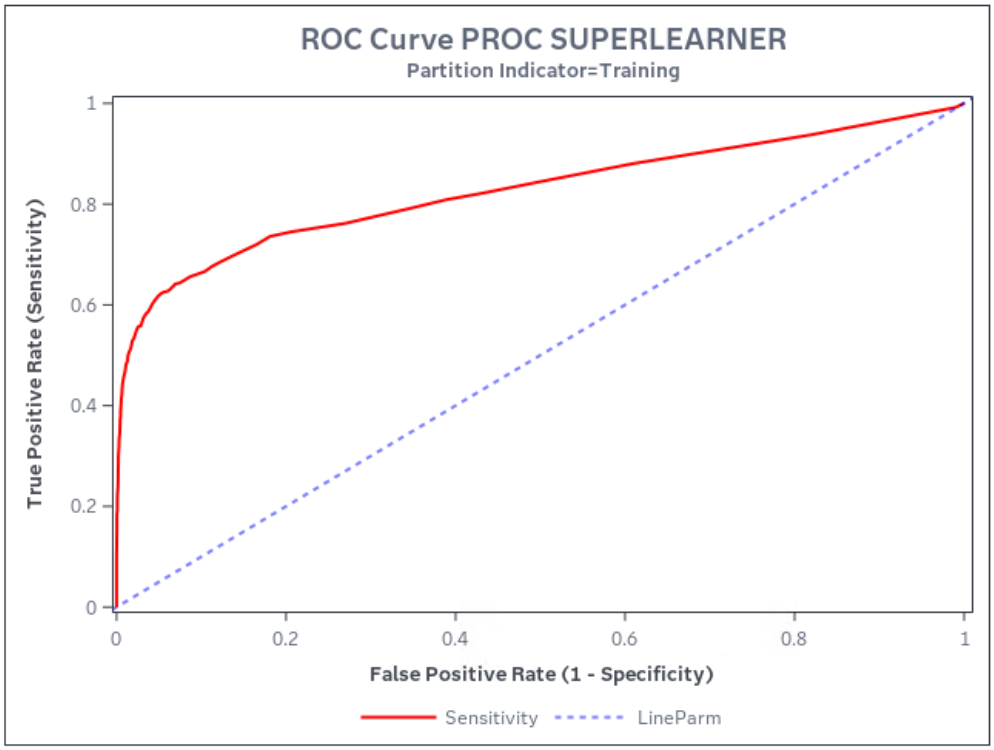
```
    by _partind_ ;
    /** Output ROC table **/
run;
ods output close;
```

The **PROC ASSESS** allows us to evaluate, compare, and validate the performance of predictive models. It helps to compute model assessment statistics such as lift, gain, ROC information by both our validation and training sets.

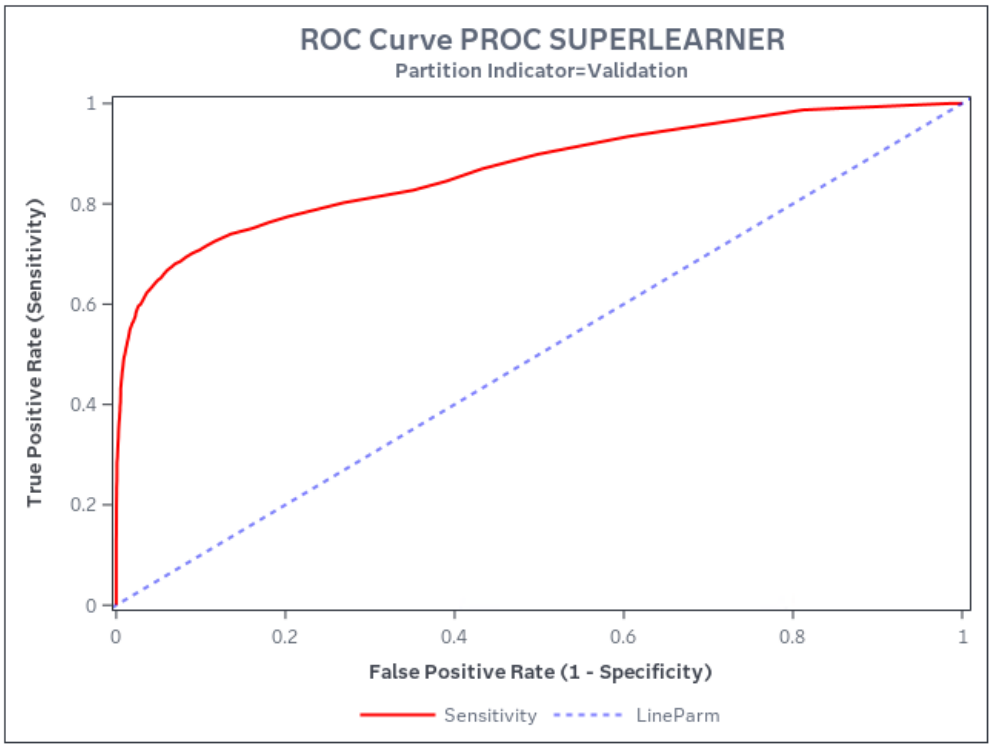
```
proc print data=out.roc_table2;
run;
```

In the above code, we use the PROC ASSESS procedure to assess the model performance and obtain the ROC Data Information. For this procedure we specify the information we want to assess. We name the data **"OUT.ROC_TABLE2"** that will provide us with ROC Curve Information need to create our ROC Curve using the PROC SGPLOT. The variable we call is **"P_churnChurn",** we call the target variable **"churn"** with the event as **"Churn"** with a level of nominal which is customers who have the probability of ending business with the telecommunication company. We separate them by the **"_PartInd_"** indicator to provide use with the ROC curve for both the Training and Validation set.

```
proc sgplot data=out.roc_table2;
    series x=FPR y=Sensitivity / lineattrs=(thickness=2 color=RED);
    by _partind_;
    lineparm x=0 y=0 slope=1 / transparency=0.5
lineattrs=(pattern=shortdash
        color=BLUE); /* diagonal */
    xaxis label="False Positive Rate (1 - Specificity)" min=0 max=1;
    yaxis label="True Positive Rate (Sensitivity)" min=0 max=1;
    title "ROC Curve PROC SUPERLEARNER";
run;
```




The Figure above provides the performance of our Training set for the PROC SUPERLEARNER model.

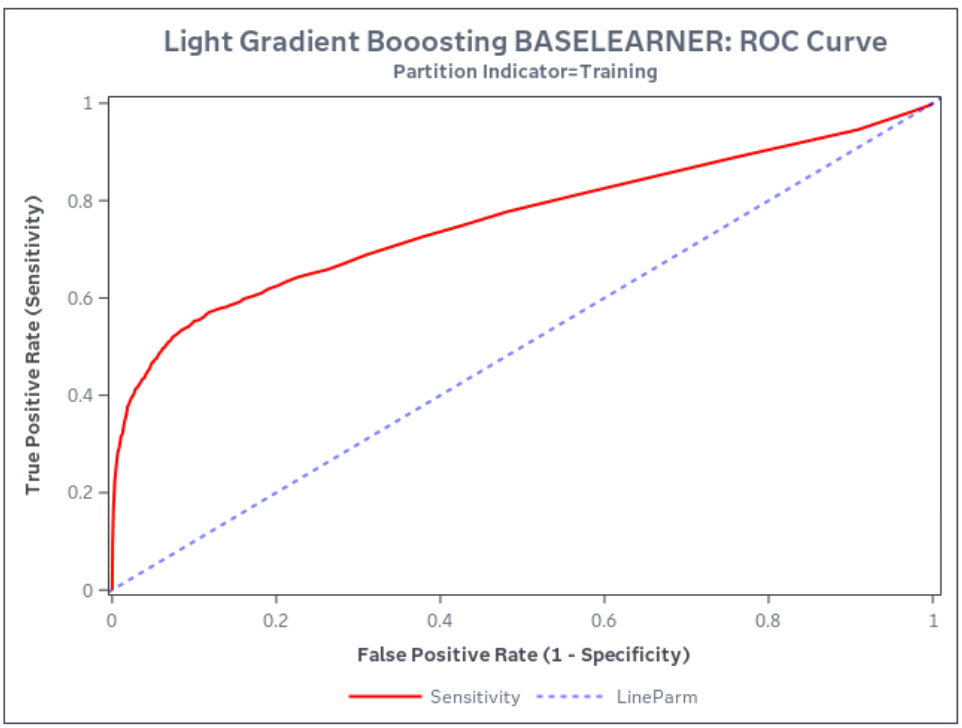


The Figure above provides the performance of our Validation set for the PROC SUPERLEARNER model.

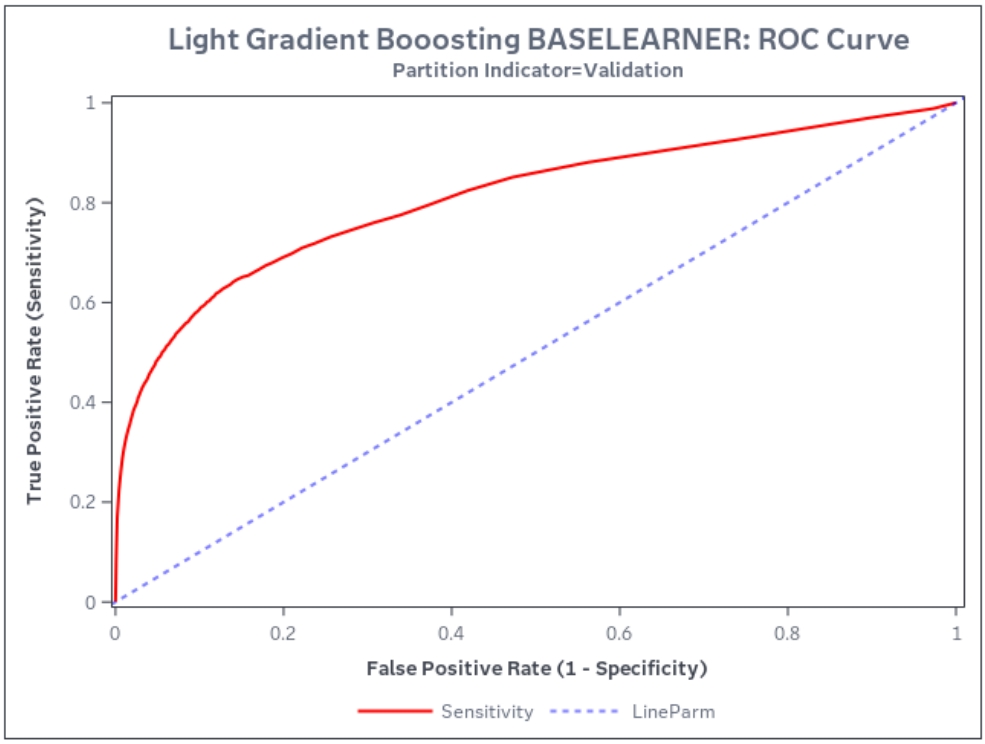
ROC Curve: BASELEARNERS

```
/*********************************************************************
****/


proc sgplot data=out.roc_table;
    series x=FPR y=Sensitivity / lineattrs=(thickness=2 color=RED);
    by _partind_;
    lineparm x=0 y=0 slope=1 / transparency=0.5
lineattrs=(pattern=shortdash
        color=BLUE); /* diagonal */
    xaxis label="False Positive Rate (1 - Specificity)" min=0 max=1;
    yaxis label="True Positive Rate (Sensitivity)" min=0 max=1;
    title "PROC SUPERLEARNER: ROC Curve";
run;

/********    ROC PROC SUPER VS BASELEARNERS   *************/;

/*********************************************************/



/***** ROC for Light-Gradient Boosting Model ****/
ods output ROCInfo=out.roc_table_LGB;

/* <----- This is the ROC Curve table output */
proc assess data=out.score_data;
    var LGB;
    target churn / event="Churn" level=nominal;
    by _partind_ ;
    /** Output ROC table **/
run;
ods output close;
```

```
proc print data=out.roc_table_LGB;
run;

proc sgplot data=out.roc_table_LGB;
    series x=FPR y=Sensitivity / lineattrs=(thickness=2 color=RED);
    by _partind_;
    lineparm x=0 y=0 slope=1 / transparency=0.5
lineattrs=(pattern=shortdash
        color=BLUE); /* diagonal */
    xaxis label="False Positive Rate (1 - Specificity)" min=0 max=1;
    yaxis label="True Positive Rate (Sensitivity)" min=0 max=1;
    title "Light Gradient Booosting BASELEARNER: ROC Curve";
run;
```



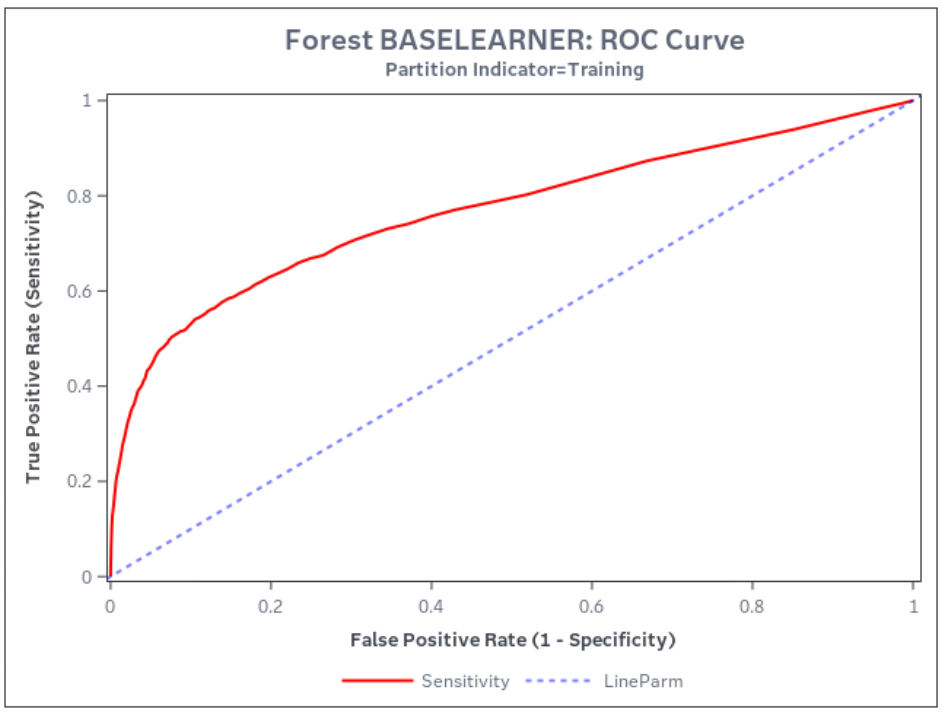

The Figure above provides the performance of our Training set for the Gradient Boosting model.



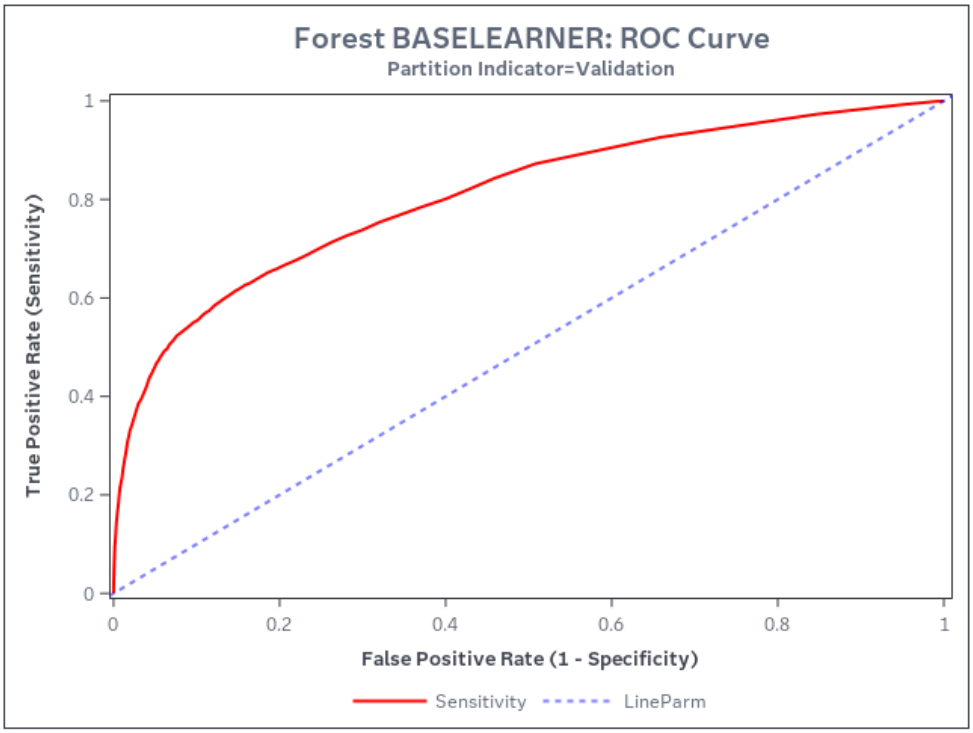

The Figure above provides the performance of our Validation set for the Gradient Boosting model.

```
/**************    ROC Curve for Forest Model    ********************/

/***************    ROC Curve Information    *************************/
ods output ROCInfo=out.roc_table_Forest;

/* <----- This is the ROC Curve table output */
proc assess data=out.score_data;
    var Forest;
    target churn / event="Churn" level=nominal;
    by _partind_ ;
    /** Output ROC table **/
run;
ods output close;

proc print data=out.roc_table_Forest;
run;
```

```
proc sgplot data=out.roc_table_Forest;
    series x=FPR y=Sensitivity / lineattrs=(thickness=2 color=RED);
    by _partind_;
    lineparm x=0 y=0 slope=1 / transparency=0.5
lineattrs=(pattern=shortdash
        color=BLUE); /* diagonal */
    xaxis label="False Positive Rate (1 - Specificity)" min=0 max=1;
    yaxis label="True Positive Rate (Sensitivity)" min=0 max=1;
    title "Forest BASELEARNER: ROC Curve";
run;
```



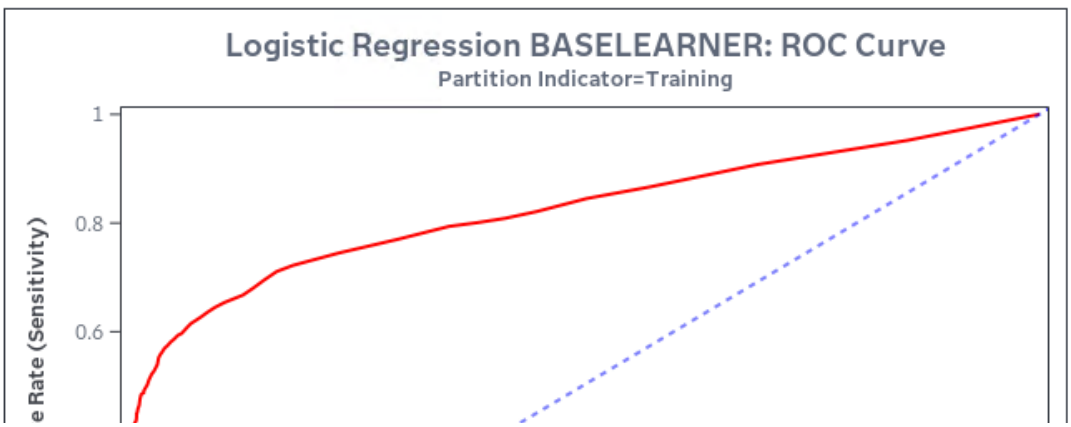

The Figure above provides the performance of our Training set for the Forest model.

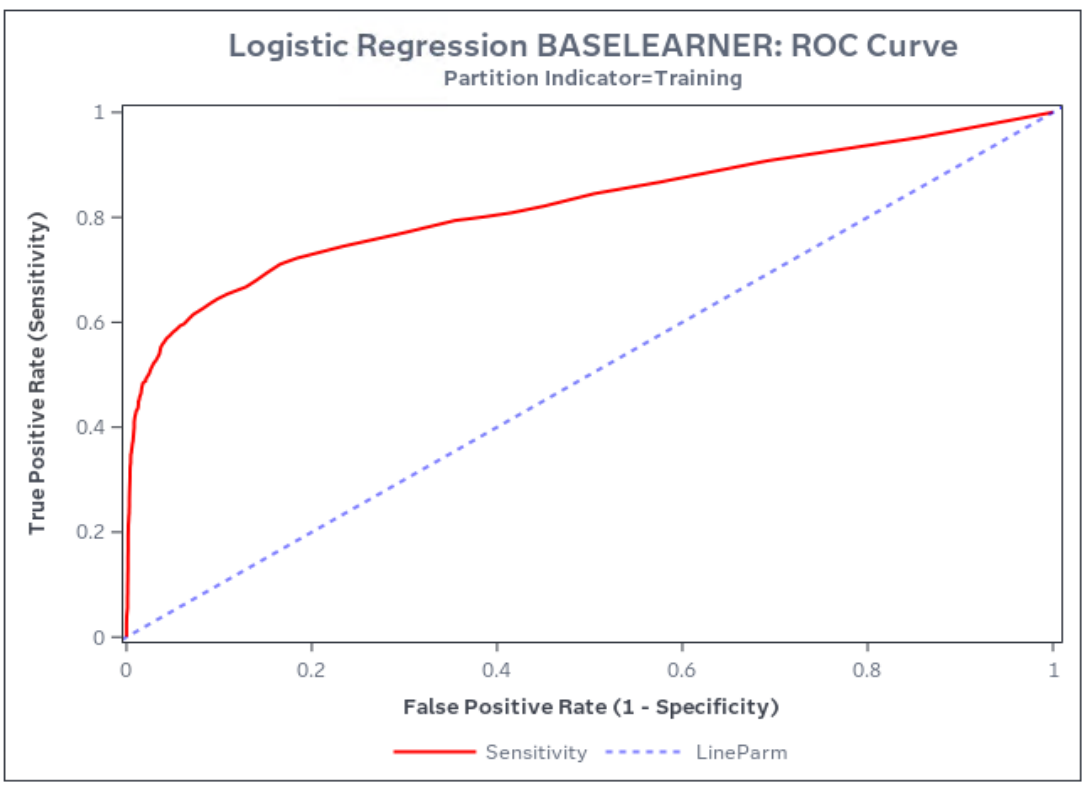



The Figure above provides the performance of our Validation set for the Forest model.

```
/***ROC CURVE for Logistic Regression Model**/


/************    ROC Curve Information   *****************/
ods output ROCInfo=out.roc_table_Logistic;

/* <----- This is the ROC Curve table output */
proc assess data=out.score_data;
    var Logistic;
    target churn / event="Churn" level=nominal;
    by _partind_ ;

run;
ods output close;

proc print data=out.roc_table_Logistic;
run;
```

```
proc sgplot data=out.roc_table_Logistic;
    series x=FPR y=Sensitivity / lineattrs=(thickness=2 color=RED);
    by _partind_;
    lineparm x=0 y=0 slope=1 / transparency=0.5
lineattrs=(pattern=shortdash
        color=BLUE); /* diagonal */
    xaxis label="False Positive Rate (1 - Specificity)" min=0 max=1;
    yaxis label="True Positive Rate (Sensitivity)" min=0 max=1;
    title "Logistic Regression BASELEARNER: ROC Curve";
run;
```

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mod

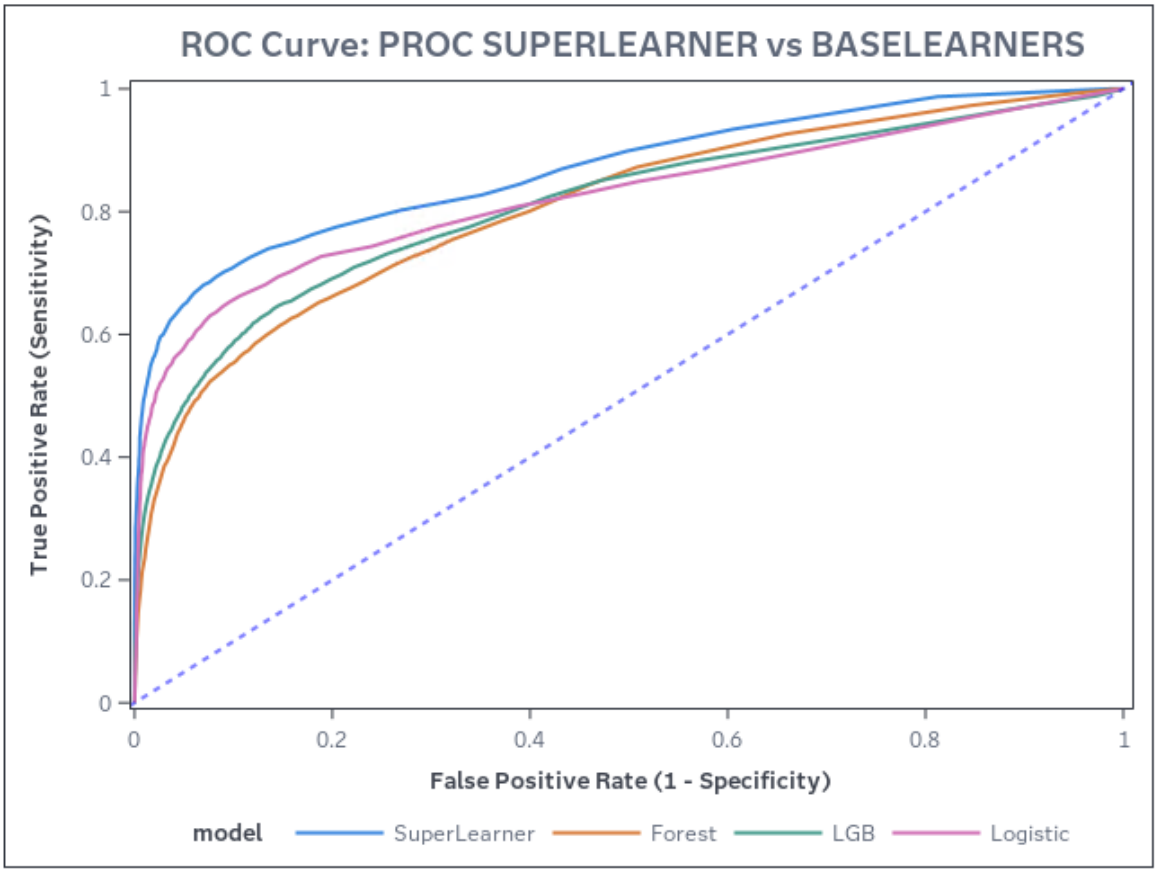
The Figure above provides the performance of our Validation set for the Logistic Regression model.

```
/*************************************************************/

/************  Create a Data step that combines all ROC tables
together **************/
data out.combined_roc_data;
    set out.roc_table out.roc_table_forest out.roc_table_lgb
out.roc_table_Logistic;
    if Variable='Forest' then model='Forest';
    else if Variable='LGB' then model='LGB';
    else if Variable='Logistic' then model='Logistic';
    else if Variable='Superlearner' then model='Superlearner';
run;

proc sgplot data=out.combined_roc_data(where=(_partind_=1));
    series x=FPR y=Sensitivity / group=model;
        /*lineattrs=(thickness=2 color=RED);*/
    lineparm x=0 y=0 slope=1 / transparency=0.5
lineattrs=(pattern=shortdash
        color=BLUE); /* diagonal */
    xaxis label="False Positive Rate (1 - Specificity)" min=0 max=1;
    yaxis label="True Positive Rate (Sensitivity)" min=0 max=1;
```

```
    title "ROC Curve: PROC SUPERLEARNER vs BASELEARNERS";
run;
```




In the figure above, we have combined the ROC table reports from each model including the overall performance of the PROC SUPERLEARNER. We overlay each model into a single ROC CURVE to show the performance of each model. We notice that the SUPERLEARNER model outperforms all the single model's performance. This ROC Curve plot shows how the PROC SUPERLEARNER takes the input predictions of each BASELEARNER to create an overall SUPERLEARNER model.

## **Optional code: PROC HPSPLIT**

The PROC HPSPLIT is high-performance procedure that builds tree-based statistical models for classification and regression. The procedure has the capabilities to produce the classification trees for categorical response, and regression trees for continuous response.

```
/**********************    Variable Importance********************/;
proc hpsplit data=ps.telecomms_part maxdepth=8;
```

```
    class churn;
    model churn=&num_inputs;
run;
```

## Full Code

```
/*********************************   LIBNAME STATEMENT     **************************/;
libname ps '/home/student/Courses/PROC_SUPERLEARNER/data';
libname out '/home/student/Courses/PROC_SUPERLEARNER/output';
/**********************************     LOAD DATA IN-MEMORY     **********************/;
data ps.telecomms;
    set ps.commsdata;
run;

/************************************** SUMMARY STATISTIC   ***************************/;
proc means data= ps.telecomms N Mean std min max skewness kurtosis descending ;
run;
/***************************** DATA ENGINEERING / FEATURE EXTRACTION ******************/;
;
/**  Handling the negative values **/;

data ps.tele_comms;
    set ps.telecomms;

 /**  The original data with negative values   **/;

 array old_limits(*) tot_mb_data_roam_curr seconds_of_data_norm lifetime_value bill_data_usg_m03
 bill_data_usg_m06 voice_tot_bill_mou_curr mb_data_usg_roamm01 mb_data_usg_roamm02 mb_data_usg_roamm03
 mb_data_usg_m01 mb_data_usg_m02 mb_data_usg_m03 calls_total calls_in_pk calls_out_pk calls_in_offpk
calls_out_offpk
 mb_data_ndist_mo6m data_device_age mou_onnet_pct_MOM mou_total_pct_MOM;

 array new_limits(*) tot_mb_data_roam_curr seconds_of_data_norm lifetime_value bill_data_usg_m03
 bill_data_usg_m06 voice_tot_bill_mou_curr mb_data_usg_roamm01 mb_data_usg_roamm02 mb_data_usg_roamm03
 mb_data_usg_m01 mb_data_usg_m02 mb_data_usg_m03 calls_total calls_in_pk calls_out_pk calls_in_offpk
calls_out_offpk
 mb_data_ndist_mo6m data_device_age mou_onnet_pct_MOM mou_total_pct_MOM;

 /**  Apply rule to set negative values to zero  **/;
 do i=1 to dim(old_limits);

 /**  If the old value is a negative value then we will set the new value to 0  **/
    if old_limits[i] < 0 then new_limits[i] = 0;

    /** If the value is greater than 0, then we keep the original value **/
    else old_limits[i] = new_limits[i];

    end;
```

```
 run;
/***********    PART 2 : TAKE THE LOG OF THE MB_DATA_Usg VARIABLES ***************/;
data ps.telecomms;
    set ps.telecomms;
    /* Original variables */
    array old_vars(*) MB_Data_Usg_M04 MB_Data_Usg_M05 MB_Data_Usg_M06
        MB_Data_Usg_M07 MB_Data_Usg_M08 MB_Data_Usg_M09;

    /* New variables */
    array new_vars(*) LOG_MB_Data_Usg_M04 LOG_MB_Data_Usg_M05
        LOG_MB_Data_Usg_M06 LOG_MB_Data_Usg_M07 LOG_MB_Data_Usg_M08
        LOG_MB_Data_Usg_M09;

   /* Apply rule */
    do i=1 to dim(old_vars);

        /* If negative → log transform */
        if old_vars[i] then new_vars[i]=log(abs(old_vars[i]));

        /* If zero or positive → keep original value
        else new_vars[i]=old_vars[i]*/;

    end;

    /* Drop variables (outside loops) */
    drop i upsell_xsell city city_lat city_long data_usage_amt
        mou_onnet_6m_normal mou_roam_6m_normal region_lat region_long state_lat
        state_long tweedie_adjusted zip_lat zip_long zipcode_primary;
run;
data ps.telecomms;
    set ps.telecomms;

    drop MB_Data_Usg_M04 MB_Data_Usg_M05 MB_Data_Usg_M06 MB_Data_Usg_M07
        MB_Data_Usg_M08 MB_Data_Usg_M09;
run;
/********* CREATE A LET STATEMENT TO STORE VARIABLE FOR ANALYSIS**************/;
;
%let num_inputs=Customer_ID lifetime_value avg_arpu_3m acct_age billing_cycle
    nbr_contracts_ltd rfm_score Est_HH_Income cs_med_home_value
    cs_pct_home_owner cs_ttl_pop cs_hispanic cs_caucasian cs_afr_amer cs_other
    cs_ttl_urban cs_ttl_rural cs_ttl_male cs_ttl_female cs_ttl_hhlds
    cs_ttl_mdage forecast_region mb_inclplan ever_days_over_plan
    ever_times_over_plan data_device_age equip_age mfg_apple mfg_samsung mfg_htc
    mfg_motorola mfg_lg mfg_nokia delinq_indicator times_delinq
    count_of_suspensions_6m avg_days_susp calls_total calls_in_pk calls_in_offpk
    calls_out_offpk calls_out_pk mou_total_pct_MOM mou_onnet_pct_MOM
    mou_roam_pct_MOM voice_tot_bill_mou_curr tot_voice_chrgs_curr tot_drpd_pr1
    bill_data_usg_m03 bill_data_usg_m06 bill_data_usg_m09 mb_data_usg_m01
    mb_data_usg_m02 mb_data_usg_m03 mb_data_ndist_mo6m mb_data_usg_roamm01
    mb_data_usg_roamm02 mb_data_usg_roamm03 tot_mb_data_curr
    tot_mb_data_roam_curr bill_data_usg_tot tot_overage_chgs
    data_prem_chrgs_curr nbr_data_cdrs avg_data_chrgs_3m avg_data_prem_chrgs_3m
    avg_overage_chrgs_3m nbr_contacts calls_TS_acct open_tsupcomplnts
    num_tsupcomplnts unsolv_tsupcomplnt wrk_orders days_openwrkorders
    resolved_complnts calls_care_acct calls_care_3mavg_acct
    calls_care_6mavg_acct res_calls_3mavg_acct res_calls_6mavg_acct
    last_rep_sat_score network_mention service_mention price_mention times_susp
    curr_days_susp pymts_late_ltd calls_care_ltd seconds_of_data_norm
    seconds_of_data_log LOG_MB_Data_Usg_M04 LOG_MB_Data_Usg_M05
    LOG_MB_Data_Usg_M06 LOG_MB_Data_Usg_M07 LOG_MB_Data_Usg_M08
    LOG_MB_Data_Usg_M09;

proc contents data=ps.telecomms;
run;
/*****************    DATA PARTITION: SPLITTING OUR DATA INTO A 70/30 SPLIT  *****************/;
proc  partition data=ps.telecomms samppct=70 seed=12345 partind;
    by churn;
    output out=ps.telecomms_part copyvars=(_all_);
run;
```

```
/****************** DATA EXPLORATION:  Check skewness of our data   *************************/;
/* Variable Importance */;
proc hpsplit data=ps.telecomms_part maxdepth=8;
    class churn;
    model churn=&num_inputs;
run;
proc contents data=ps.telecomms_part out=vars(keep=name) noprint;
run;
/************************BUILDING THE PROC SUPERLEARNER MODEL ****************************/;
proc superlearner data=ps.telecomms_part seed=12345;
    target churn / level=interval;
    input &num_inputs / level=interval;
    baselearner 'LGB' lightgradboost(leafsize=50 numbin=100);
    baselearner 'Forest' forest( maxdepth=15 inbagfraction=0.7 minleafsize=15);
    baselearner 'GB' gradboost(learningrate=0.01 maxdepth=15 minleafsize=5);
    baselearner 'DT' treesplit( criterion=gini minleafsize=15);
    crossvalidation kfold=5;
    output out=ps.predict_output copyvar=churn;
run;
```